

Q6. Sensitivity Analysis for Potential MNAR Mechanisms (Continuous Outcome)

The validity of likelihood-based inference for the continuous BPRS outcome relies on the Missing At Random (MAR) assumption. Although exploratory analyses and subsequent modeling steps provided support for MAR, this assumption cannot be verified empirically. It is therefore important to assess the robustness of the substantive conclusions to plausible violations of MAR. To this end, a sensitivity analysis was conducted under a Missing Not At Random (MNAR) framework for the continuous outcome.

MNAR Sensitivity Analysis Strategy

A pattern-mixture, delta-adjustment approach was adopted to explore departures from MAR. Missing BPRS values were first multiply imputed under MAR using a fully conditional specification with a linear regression model including time, quadratic time, and baseline covariates. MNAR scenarios were then generated by applying a constant shift δ to imputed values corresponding to originally missing observations, while leaving observed values unchanged.

Positive values of δ represent scenarios in which unobserved BPRS values are systematically higher than predicted under MAR, whereas negative values represent systematically lower unobserved values. A small but informative set of increasingly extreme shifts ($\delta = -10, -5, 0, 5, 10$) was considered. For each MNAR scenario, the final linear mixed-effects model with a patient-specific random intercept and random slope for time was refitted, and fixed-effect estimates were pooled across imputations using Rubin's rules.

Results Across MNAR Scenarios

Table 10 summarizes the pooled fixed-effect estimates for selected covariates across the different δ values.

Table 10. Pooled fixed-effect estimates across MNAR shift parameters (δ).

Effect	Est_-10	Est_-5	Est_0	Est_5	Est_10	_LABEL_	P_-10	P_-5	P_0	P_5	P_10
abpet0	1.870616	2.315527	2.741717	3.158370	3.584259	Pr > t	<.0001	<.0001	<.0001	<.0001	<.0001
adl	0.073830	0.044875	0.016047	-0.012442	-0.040789	Pr > t	0.1866	0.3688	0.7337	0.7962	0.4394
age	1.879865	1.895869	1.912678	1.930614	1.948744	Pr > t	<.0001	<.0001	<.0001	<.0001	<.0001
bmi	0.050352	0.095279	0.138699	0.181493	0.225143	Pr > t	0.3311	0.0408	0.0015	<.0001	<.0001
taupet0	-3.130042	-1.721615	-0.287783	1.046901	2.276559	Pr > t	0.0073	0.1105	0.7815	0.3168	0.0384
time	8.536325	8.338857	8.141389	7.943921	7.746453	Pr > t	<.0001	<.0001	<.0001	<.0001	<.0001
time2	-0.048245	-0.039647	-0.031049	-0.022450	-0.013852	Pr > t	0.0046	0.0123	0.0424	0.1501	0.4118

The linear time effect remains large and positive for all scenarios, and its statistical significance is preserved throughout the entire range of MNAR shifts. The quadratic time effect remains negative across all δ values, with some attenuation in magnitude under extreme shifts, but without any reversal in sign or qualitative interpretation. Baseline covariate effects show only minor fluctuations and do not become substantively important under any of the considered MNAR scenarios.

To complement the numerical results, Figure 3 presents the pooled estimates of the linear and quadratic time effects as a function of δ .

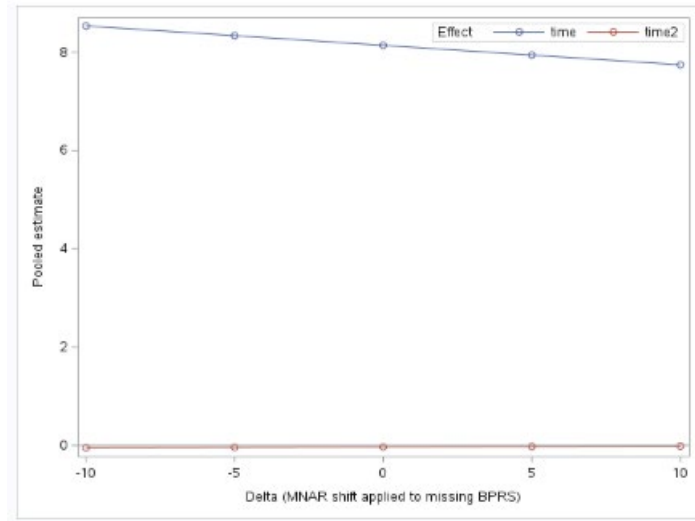


Figure 3: Sensitivity of pooled time and time² estimates to MNAR shifts (tipping-point plot)

The figure clearly shows that the estimated time trajectory is highly stable: the linear time effect varies only modestly as δ changes, and the quadratic term remains close to zero and negative. No “tipping point” is observed within the investigated range, meaning that none of the substantive conclusions change even under fairly extreme MNAR assumptions.

Interpretation and Implications

This sensitivity analysis indicates that the main conclusions regarding the longitudinal evolution of BPRS are robust to moderate and even substantial departures from MAR. In particular, the strong time trend and its non-linear shape persist across all MNAR scenarios considered. Only under unrealistically large shifts would one expect meaningful changes in inference, providing a clear and interpretable tipping-point perspective.

These findings strengthen confidence in the likelihood-based LMM results presented earlier. While MNAR mechanisms can never be ruled out empirically, the present analysis shows that plausible MNAR deviations would not materially alter the substantive interpretation of disease progression in this dataset.

Conclusion

In summary, a delta-adjusted MNAR sensitivity analysis demonstrates that inference on the continuous BPRS outcome is highly robust. The consistency of results across MAR-based modeling and MNAR sensitivity scenarios supports the credibility of the primary conclusions and suggests that potential violations of MAR are unlikely to have a meaningful impact on the substantive findings.